Hidden World of Engineering: An Analysis of the FIRST Robotics Discourse Community

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At some school orientations, mostly secondary schools, there are the standard extracurricular booths showcasing athletics and standard hobby clubs, which can frequently blend together in terms of activities that do not really stick out for some. However teams associated with FIRST Robotics tend to stand out by presenting some advanced technical tools, some form of digital media, and a potentially operational robotic unit. Though casual observers likely tend to misclassify FIRST Robotics as a standard recreational club, it does function as a distinct discourse community under the framework established by a linguist by the name of John Swales. Members of this community typically rely on a network of specialized writing genres – ranging from technical documentation to collaborative internal channels – alongside a specific shared lexicon to achieve their team's engineering goals under shorter competitive deadlines. This profile analyzes how the community structures communication to bridge a gap between novice students, experts in the forms of mentors, and external stakeholders who fund the community.

As per John Swales' First Criterion, the shared goal of this community is quite twofold: using the engineering process to manifest a functional robot under strict rules, and upholding the core philisophy of FIRST, of competing hard but helping other teams, calling it Gracious Professionalism. This philisophy dictates that teams should compete intensely while actively assisting their opponents outside of competition matches. This collaborative dynamic is certified by the competition matches of the FIRST Robotics Competition and First Tech Challenge, where an opposing team in one match may just be a critical alliance partner in the next. In terms of audiences these members come into contact with, they typically talk to mentors and other sub-teams internally, while externally they communicate with corporate sponsors for funding their team, alongside talking with competition judges to try and win

awards to increase their chances of making it to larger competitions, with the eventual goal of making it to the international competition to compete with others who also made it there. Showing that they have managed to accomplish skills that they can use later in life whether it be mechanical engineering, or software development.

When a student decides to join a FIRST Robotics team, they typically have to expunge some money for a pay to play operational fee alongside having one of their parents or legal guardians sign a consent agreement in order to operate industrial tools. Once inside the community, members must assist in producing specific documents, primarily the engineering notebook. The engineering notebook is a highly structured multiple page record containing Computer Aided Designs, programming logic, prototyping logs, alongside math calculations. The Engineering Notebook's purpose is to provide objective evidence of the team's design journey for tournament judges.

Swales' framework requires a discourse community to provide a specific lexis, a unique vocabulary, FIRST Robotics also uses its own unique language, terms, and acronyms that are used by the various teams that are part of this discourse community. Some examples include the "Autonomous Period", a section of a competition match dedicated to the robot running entirely on pre-written code without human interaction, alongside the "Teleoperated Period", often simplified down to Teleop, where the drive team subteam of a team manually controls a robot via joysticks and buttons. There is also some shorthand phrases that members of this community use, such as "COTS", being short for Commercial Off The Shelf parts, or "The Pit", which is a small quite high-stress garage space assigned to teams at competitions. These shorthand statements allow for mechanical builders, software programmers, and the drive teams to share large amounts of technical data near instantaneously without wasting words and breath. While technical sub-teams use shorthand like "COTS" and "Teleop" internally, other subteams, primarily the business sub-team, must entirely translate such terms into plain English when

writing grant applications or speaking to corporate sponsors who currently do not know anything about robotics.

Although communication is a large part of FIRST Robotics, communication often faces quite severe friction during actual competitions, as at a competition, typically there is quite a lot of noise, such as people cheering for other members of their team, music is blaring in between matches, and pit announcements often tend to interrupt the attention spans that were initially focused on a robot. Not to mention that between the match a team just finished and their next match, they typically have less than ten or fifteen minutes between their matches to completely diagnose any hardware failures or software bugs on top of the noise causing communication issues. Contributing to this document serves as an important milestone for students. Learning how to properly format such entries and accurately recording mechanical failures ensures a novice member is fully integrated into the team's professional work. Through such an iterative writing process, students stop acting as though they're passive observers and beome active, contributing members of the engineering group.

Primary research gathered from an active community member highlights that such communication hurdles exist not only during the build season, but also at live competitions. The interviewee noted that some students not familiar with robotics can get confused by some of the terms and part names at the beginning of the season, and also running into issues with multiple students wanting to drive the robot at these competitions. This insight reveals a quite significant internal challenge regarding the entry level of the community. New members face a quite steep learning curve trying to decipher established lexicon, creating a barrier to entry that requires deliberate mentoring. Furthermore, a conflict of multiple students wanting to occupy a coveted "driver" role necessitates quite clear, structured rules of communication and operational hierarchy within the team to resolve any interpersonal tension and to maintain focus during high pressure events.Other knowledge gathered from this member also mentions that they believe that students involved learning new skills and everyone being involved during the

season is more of a success than winning trophies, alongside them enjoying learning new information along with students and helping them develop skills to be successful in the future. This is what keeps this member active in this community.

Overall, FIRST Robotics is much more than an after school hobby, rather it is a fully functioning technical discource community. By navigating complex textual genres like the Engineering Notebook, gaining insight into a new lexicon, and overcoming stressful communication challenges, students are trained to act like professional engineers. These communication conventions appear to successfully bridge the gap between classroom theory and real world engineering execution.

